

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2221

Girard 100 kV
November 25, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2221. The Interconnection Customer proposes a 28.6 MW battery energy storage facility interconnecting at the Girard 100 kV substation in the Ohio area. The request was received on November 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 445 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,598,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MAXWELL – CORONA 100 kV
Contingency: CIMARRON RIV – RAVENNA 100 kV
Loading: 130.4% (Rating: 123 MVA)
DFAX: 5.4%
Pre-project loading: 129.1%
2. Facility: HOLLYBROOK – MAXWELL 100 kV
Contingency: RAVENNA – CIMARRON RIV 100 kV
Loading: 126.4% (Rating: 115 MVA)
DFAX: 5.4%
Pre-project loading: 125.1%
3. Facility: TUTHILL – HOLLYBROOK 100 kV L
Contingency: RAVENNA – CIMARRON RIVER 100 kV
Loading: 124.3% (Rating: 102 MVA)
DFAX: 5.4%
Pre-project loading: 122.8%
4. Facility: CRANBROOK – MANTUA 100 kV
Contingency: MANTUA – CAWELO C 100 kV
Loading: 108.5% (Rating: 103 MVA)
DFAX: 7.6%
Pre-project loading: 106.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MAXWELL – CORONA 100 kV	Rebuild line with higher-capacity conductor	\$3,631,000	100% (\$3,631,000)
MAXWELL – HOLLYBROOK 100 kV	Rebuild line with higher-capacity conductor	\$2,070,000	100% (\$2,070,000)
TUTHILL – HOLLYBROOK 100 kV	Rebuild line with higher-capacity conductor	\$3,702,000	100% (\$3,702,000)
CRANBROOK – MANTUA 100 kV	Reconductor line	\$6,195,000	100% (\$6,195,000)
	Total allocated to project		\$15,598,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.